

MAJAVATN V, Norway

WMO: 011320

Lat: 65.1661N

Lon: 13.3667E

Elev: 1047

StdP: 14.15

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.5	-9.2	-20.9	1.8	-15.4	-14.1	2.7	-8.9	22.6	20.6	19.1	23.1	2.0	20	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	76.2	60.0	71.9	58.3	67.9	56.3	62.0	73.3	59.5	69.1	57.5	66.1	5.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.7	73.7	65.8	55.1	67.2	63.2	53.0	62.1	61.6	28.2	73.4	26.5	69.3	25.2	66.0	70.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	13.4	11.5	-18.3	80.6	5.6	4.7	-22.4	84.0	-25.7	86.7	-28.8	89.3	-32.9	92.7
			-18.7	63.1	5.6	3.6	-22.7	65.7	-26.0	67.8	-29.1	69.8	-33.1	72.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.4	19.5	19.8	24.0	33.4	41.4	49.9	56.9	53.7	47.5	37.0	28.9	23.3
	DBStd	15.49	12.12	12.41	10.24	6.30	6.35	6.22	6.51	5.04	5.55	6.13	9.76	11.26
	HDD50	5456	946	847	805	498	282	76	10	19	109	405	632	827
	HDD65	10464	1411	1267	1270	948	733	453	264	350	525	869	1082	1292
	CDD50	483	0	0	0	0	14	74	224	136	34	1	0	0
	CDD65	15	0	0	0	0	0	1	13	1	0	0	0	0
	CDH74	227	0	0	0	0	6	26	184	11	0	0	0	0
	CDH80	40	0	0	0	0	0	2	38	0	0	0	0	0
Wind	WSAvg	4.8	5.6	5.8	5.4	4.5	4.3	4.1	3.9	4.0	4.5	4.9	5.0	5.5
Precipitation	PrecAvg	55.0	5.8	5.6	5.6	3.4	2.5	3.1	3.4	3.5	5.4	5.7	4.9	6.2
	PrecMax	73.6	12.9	13.4	11.5	8.3	4.4	6.8	8.5	6.5	15.0	11.4	10.9	12.1
	PrecMin	39.6	0.4	0.4	0.9	1.0	0.7	0.6	1.5	0.7	1.6	2.0	0.3	0.6
	PrecStd	8.6	3.4	3.4	3.1	1.9	0.8	1.4	1.5	1.3	3.0	2.4	3.0	3.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.0	40.8	42.5	56.7	71.4	77.3	84.9	75.5	67.2	55.4	47.6	41.7
		MCWB	37.3	37.8	36.1	44.4	54.1	58.3	65.1	60.2	55.2	50.1	45.3	39.5
	2%	DB	37.5	38.7	40.0	51.0	64.2	71.5	79.3	71.8	62.3	50.9	44.5	39.5
		MCWB	35.4	36.4	35.7	41.5	51.0	55.8	61.8	58.7	53.3	45.8	41.6	37.0
	5%	DB	35.7	37.2	38.4	45.9	58.6	66.4	74.4	68.2	59.2	48.2	41.0	37.5
		MCWB	33.3	34.8	35.0	38.5	47.6	53.7	60.4	57.2	52.3	44.1	38.1	34.8
	10%	DB	33.8	35.1	36.7	42.4	53.3	62.1	70.1	64.5	56.5	45.4	39.1	35.9
		MCWB	32.0	32.5	33.7	36.5	45.1	51.8	58.1	55.3	50.4	42.1	36.6	33.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.3	38.3	38.4	44.9	55.3	60.4	67.8	63.2	56.7	49.9	45.6	40.1
		MCDB	39.7	39.9	40.6	55.9	69.1	73.4	82.3	72.0	63.0	54.4	47.3	41.3
	2%	WB	35.8	36.6	36.7	41.9	52.1	57.4	63.9	60.4	54.7	46.4	41.8	37.4
		MCDB	37.2	38.7	39.1	50.1	61.8	68.4	75.5	68.3	60.9	50.0	44.3	39.1
	5%	WB	33.6	35.1	35.3	39.1	48.4	55.1	61.1	58.2	52.7	44.4	38.5	35.0
		MCDB	35.0	37.0	37.9	44.4	58.0	64.5	72.3	65.9	58.6	47.6	40.3	36.9
	10%	WB	32.1	32.7	33.8	37.3	45.2	52.7	58.9	56.2	50.9	42.5	36.7	33.5
		MCDB	33.7	34.7	36.6	41.6	52.5	60.8	68.4	63.8	55.6	45.3	38.8	35.5
Mean Daily Temperature Range	5% DB	MDBR	11.6	13.3	16.1	15.2	15.7	15.4	16.7	16.5	13.0	9.8	8.5	11.0
		MCDBR	8.9	8.5	11.6	20.2	26.5	25.8	25.7	24.4	18.9	13.0	7.6	9.0
		MCWBR	8.3	7.2	8.8	12.0	14.6	12.2	11.4	12.4	11.3	9.1	6.5	7.9
	5% WB	MCDBR	8.6	8.8	9.9	17.8	24.9	22.3	23.3	20.9	16.9	11.0	7.6	9.0
		MCWBR	8.1	7.8	8.0	11.0	14.1	10.8	11.1	11.7	10.6	8.3	6.6	8.1
Clear-Sky Solar Irradiance	taub		0.194	0.265	0.297	0.326	0.333	0.333	0.342	0.329	0.309	0.285	0.353	0.198
	taud		1.923	2.142	2.183	2.219	2.348	2.472	2.478	2.503	2.546	2.379	2.702	1.935
	Ebn at Noon		97	196	245	266	277	280	273	266	248	194	95	23
	Edh at Noon		7	18	28	34	33	30	29	26	20	14	7	2
All-Sky Solar Radiation	RadAvg		30	161	571	1123	1436	1498	1417	1103	590	241	55	9
	RadStd		3	30	93	120	104	185	175	124	79	37	8	1

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page